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## Unit – 1 : Session – 9 : SLO – 2 | Logical expressions and circuits

### ■ Part B – Multiple Choice Questions

1. C — All inputs are 1
2. B — OR
3. B — 1
4. B —  $(A \cdot B)'$  (AND followed by NOT = NAND)
5. C — 4 rows ( $2^2$ )

### ■ Part A – Match the Following

1. AND gate — c) Multiplication of inputs ( $A \cdot B$ )
2. OR gate — a) Output is 1 if at least one input is 1
3. NOT gate — b) Output is the opposite of the input
4. Truth Table — d) Tabular representation of inputs and outputs
5. Logic Circuit — e) Diagram showing gates connected together

### ■ Part C – Activity Questions

1. Truth table for  $F = A \cdot B + C$  (8 rows): 000→0, 001→1, 010→0, 011→1, 100→0, 101→1, 110→1, 111→1.
2. Circuit: inputs A and B feed a 2-input AND gate; the AND output and input C feed a 2-input OR gate; OR output = F.
3. Real-life example — car seatbelt warning: buzzer = (ignition ON) AND (seatbelt NOT fastened) OR (door OPEN). Uses AND (ignition·belt■), NOT (belt status inverted), and OR (adds door condition) together.